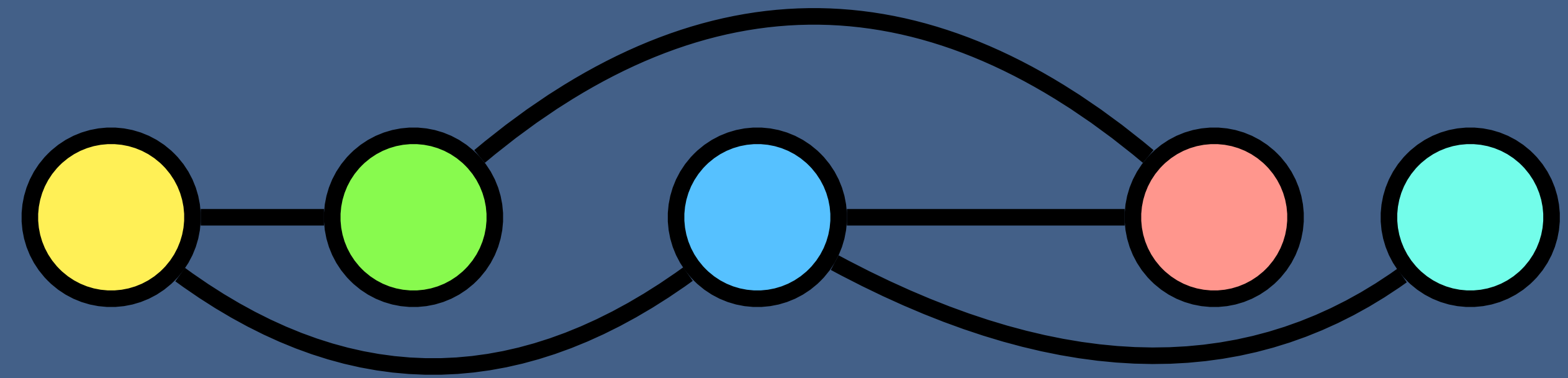
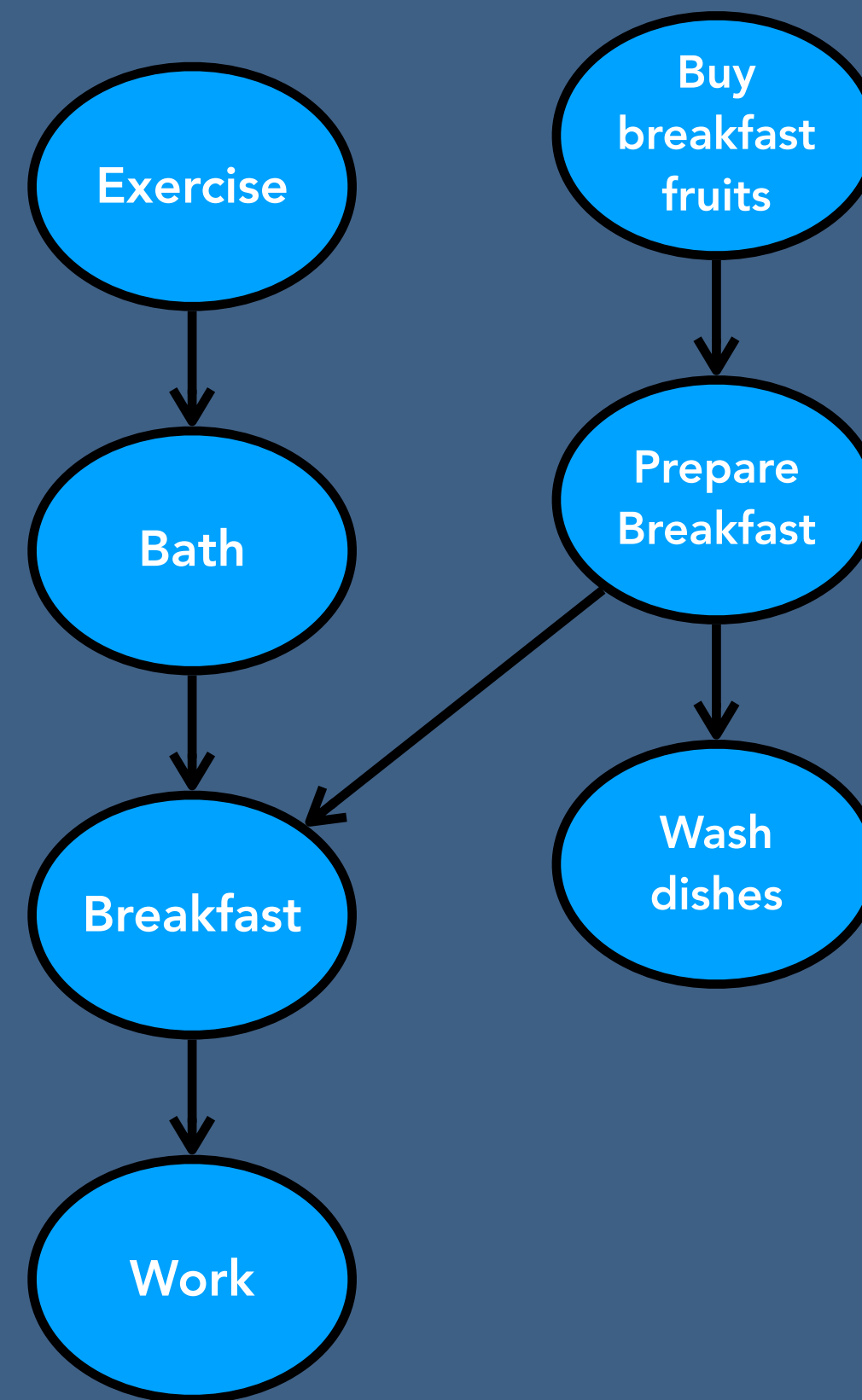


Topological Sort



Topological Sort

Topological Sort: Sorts given actions in such a way that if there is a dependency of one action on another, then the dependent action always comes later than its parent action.



Topological Sort Algorithm

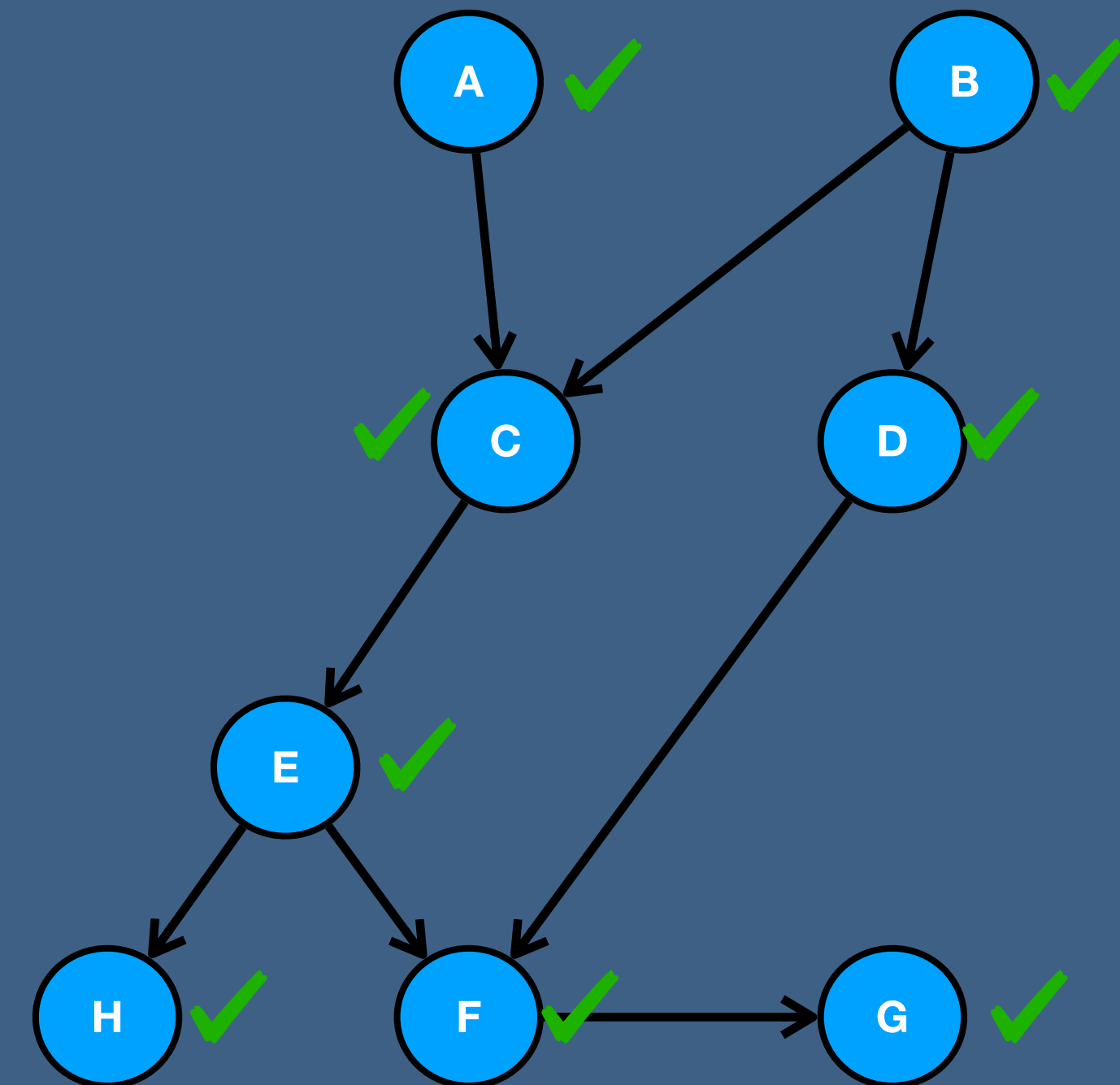
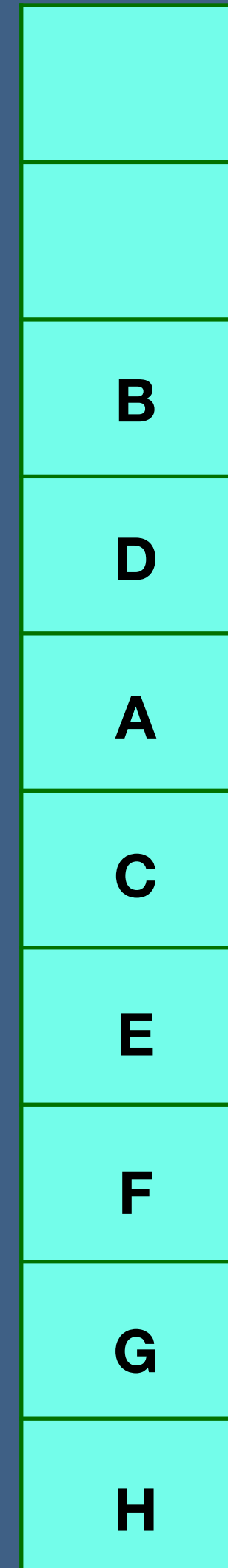
```
if a vertex depends on currentVertex:  
    Go to that vertex and  
    then come back to currentVertex  
else  
    Push currentVertex to Stack
```

A B C D E H F G

B A C D E H F G

B D A C E F G H

Stack



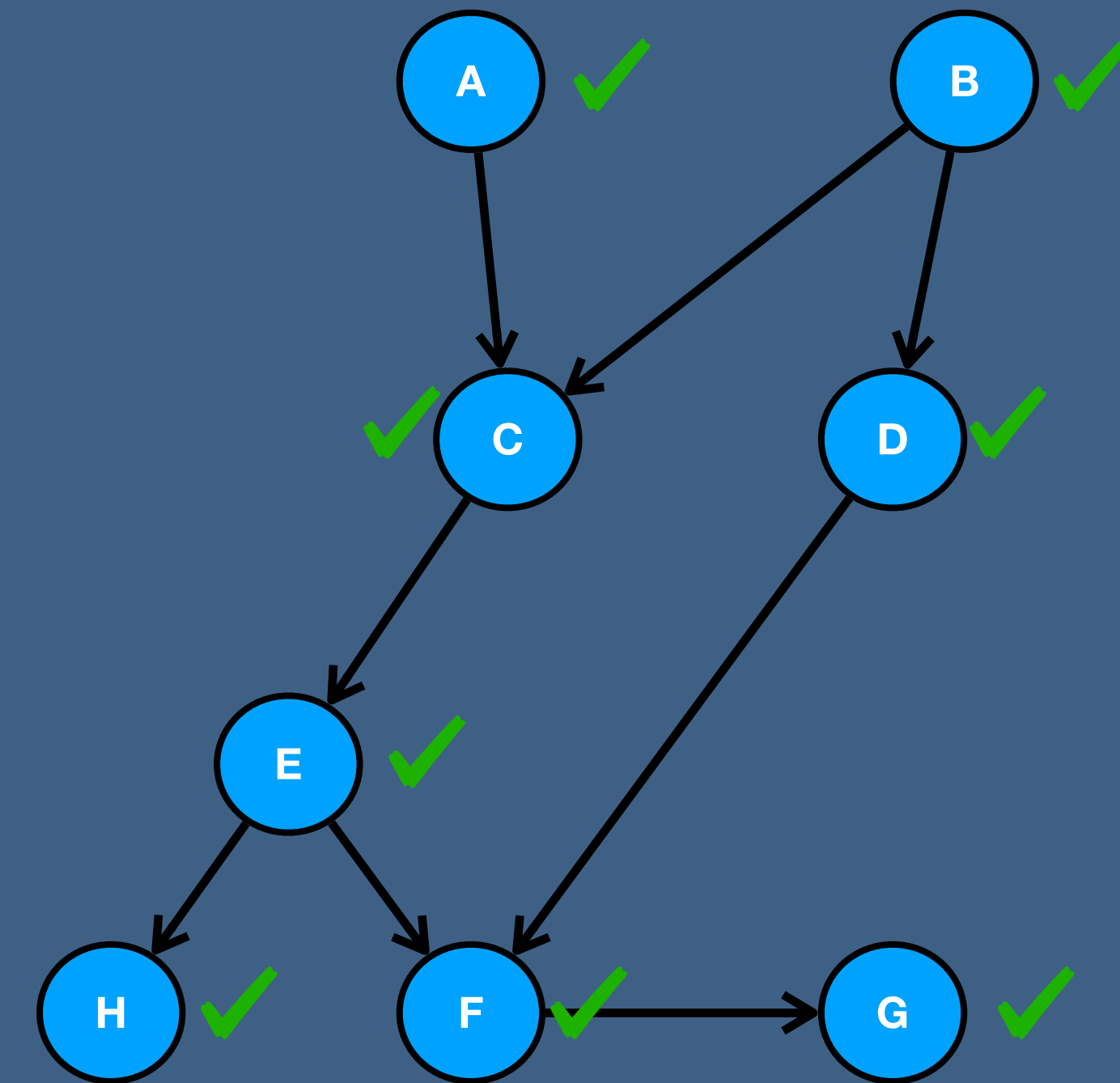
Time and Space Complexity of Topological Sort

```
topologicalSort(G)
  for all nodes
    if vertex not visited
      topologicalVisit(node)
```

} $O(V)$

```
topologicalVisit(node)
  for each nodes
    if not visited
      topologicalVisit(neighbor)
```

} $O(E)$



Time Complexity : $O(V+E)$

Space Complexity : $O(V+E)$

